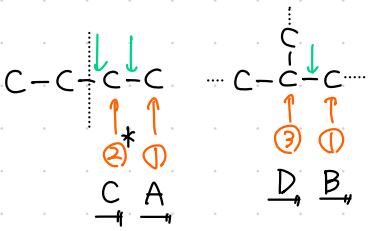
3-1 $C_4H_{10}O$ $U=1$ { アルコール / エーテル



アルコール A① B① C* D②

Ox E(アルデヒド) F(アルデヒド) G X

b.p. A > B

問1

A : $CH_3-CH_2-CH_2-CH_2-OH$

B : $CH_3-CH(CH_3)-CH_2-OH$

C : $CH_3-CH_2-CH(OH)-CH_3$

D : $CH_3-C(CH_3)(OH)-CH_3$

問2 ヒドロキシ基を持ち，分子間で水素結合を形成するため．

問3

E : $CH_3-CH_2-CH_2-CHO$

F : $CH_3-CH(CH_3)-CHO$

G : $CH_3-CH_2-CO-CH_3$

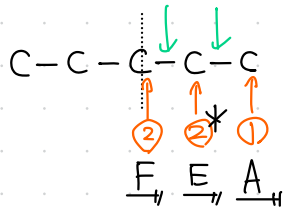
C, G

問4. $CH_3-CH_2-O-CH_2-CH_3$

$CH_3-CH_2-CH_2-O-CH_3$

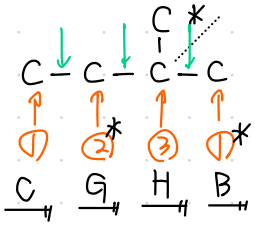
$CH_3-CH(CH_3)-O-CH_3$

3-② $C_5H_{12}O$ $U=0$ { アルコール / エーテル



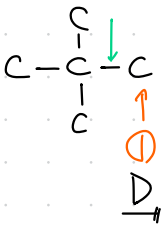
アルコール ① A ① B* ① C ① D ② E* ② F ② G* ③ H

Ox ○ ○ ○ ○ × (アルデヒド / ケトン)



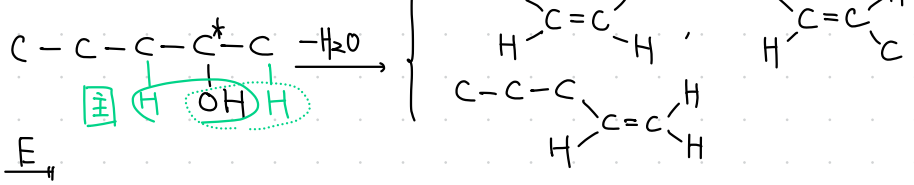
I_2 + NaOH, $-H_2O$: A' + F' (同一C骨格, ×) シス・トランスなし

$+H_2$ → 同一

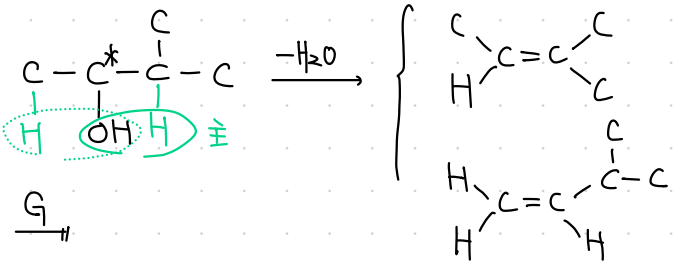


◆「脱水して～」「H_2付加して～」} 同一化合物 ⟶ 同一C骨格

〈2級 E*, G*について〉



E



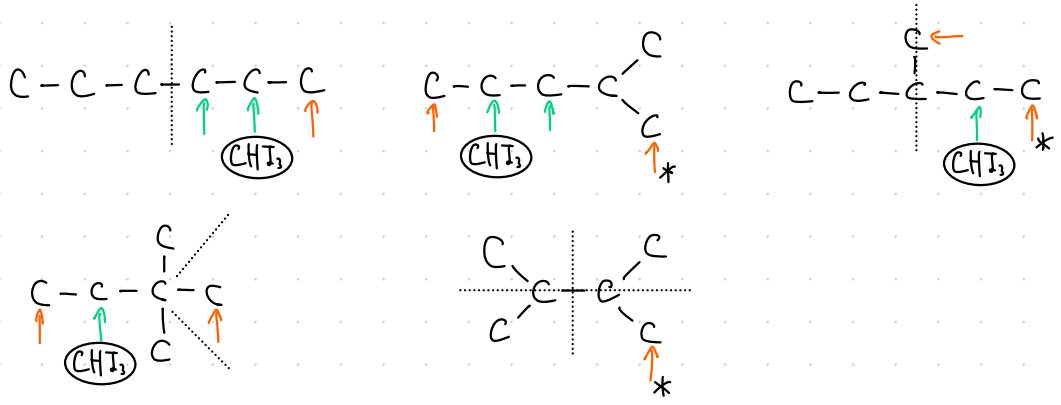
G

(2) 6 (立体異性：7)

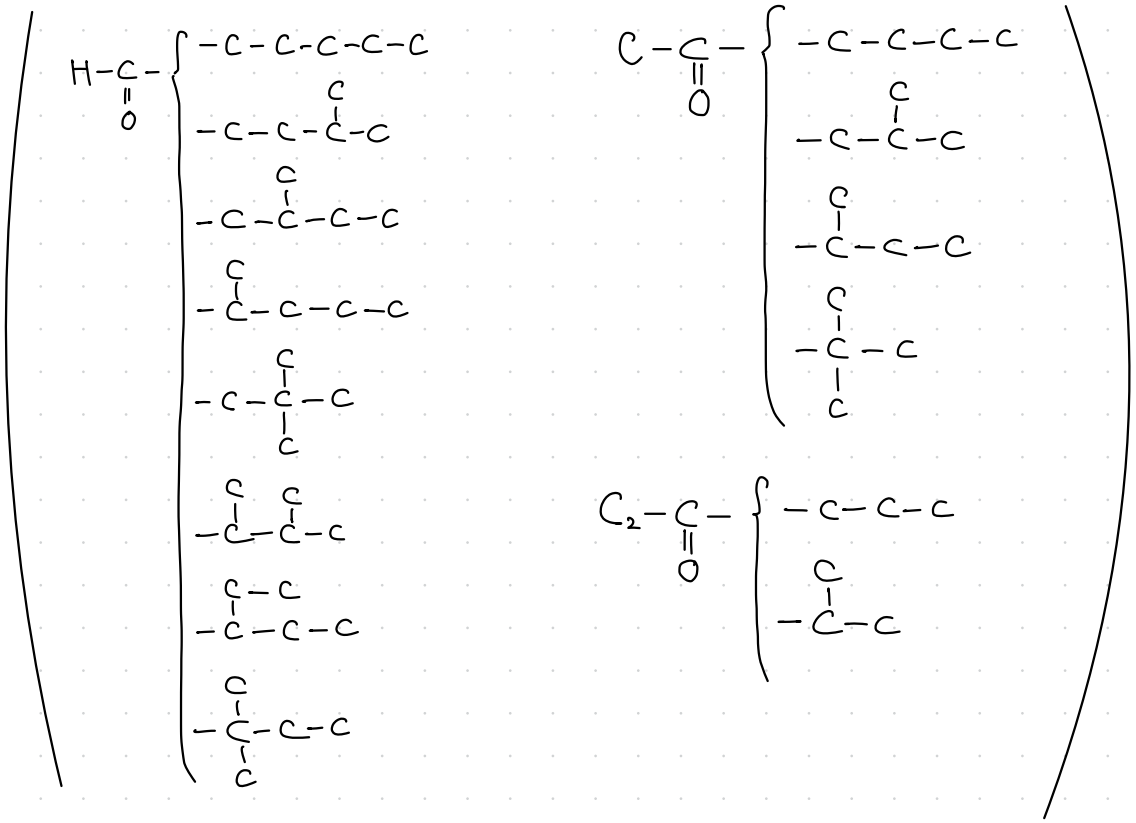
3 − ③

← : =O (アルデヒド)
← : =O (ケトン)

問1

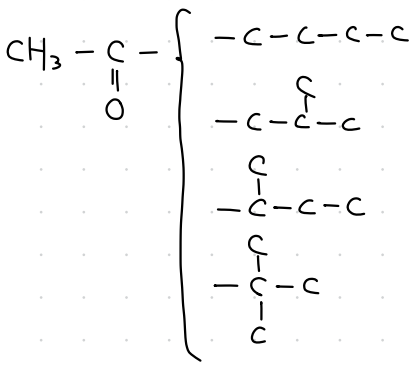


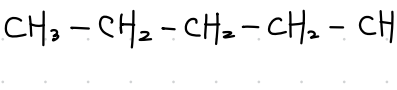
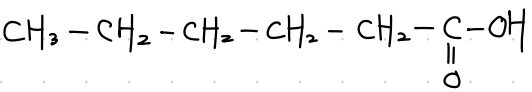
(a) 8　　(b) 6



問2. 3

問3 4



問4 X  　Y 

3－④

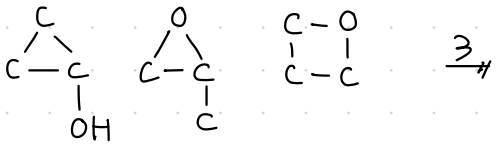
問1　A　$CH_2=CH-CH_2-OH$

B　$CH_2=CH-O-CH_3$

C　$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H$

D　$CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

問2



3

3－⑤　$C_4H_8O_2$　$U=1$　$\sim\sim -\overset{\overset{O}{\|}}{C}-O-$ ×1.

$H-COO-C_3$ ⟨ D: $-\overset{\overset{O}{\|}}{C}-C-C$ $\xrightarrow{+H_2O}$ Red: $HCOOH$ + $C-C-C-OH$

A: $-\overset{\overset{C}{|}}{C}-C$ ⟶ Red: $\overset{\overset{O}{\|}}{C}$ + $C-\overset{\overset{OH}{|}}{C}-C$ CHI_3

C: $\overset{\overset{O}{\|}}{C}-COO-C_2$ ⟶ $C-COOH$ + $C-C-OH$ CHI_3

B: $C_2-\overset{\overset{O}{\|}}{C}-COO-C$ ⟶ $C-C-COOH$　$C-OH$

b.p.　$C-OH$ (Bから) < $C-C-OH$ (Cから) < $C-\overset{\overset{OH}{|}}{C}-C$ (2級, Aから) < $C-C-C-OH$ (1級, Dから)

異性体

小 ← 分子量 → 大

A：$H-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{CH_3}{|}}{CH}-CH_3$　ギ酸イソプロピル

B：$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-O-CH_3$　プロピオン酸メチル

C：$CH_3-\overset{\overset{O}{\|}}{C}-O-CH_2-CH_3$　酢酸エチル

D：$H-\overset{\overset{O}{\|}}{C}-O-CH_2-CH_2-CH_3$　ギ酸プロピル

3－6

問1　$w_C = 264\,g \times \frac{C\,(12)}{CO_2\,(44)} = 72\,g$

$w_H = 90\,g \times \frac{2H\,(2)}{H_2O\,(18)} = 10\,g$

$w_O = 114 - (72 + 10) = 32\,g$

$C : H : O = \frac{72\,g}{12\,g/mol} : \frac{10\,g}{1.0\,g/mol} : \frac{32\,g}{16\,g/mol} = 3 : 5 : 1$

$(C_3H_5O)_n = 57n = 228 \quad n = 4$　　$C_{12}H_{20}O_4$

＜別解＞　燃焼あり，分子量既知のとき

$C_xH_yO_z \xrightarrow{O_2} xCO_2 + \frac{y}{2}H_2O$

$C_xH_yO_z : CO_2 = 1 : x$ より，
(288)　(44)

$\frac{114\,g}{228\,g/mol} \times x = \frac{264\,g}{44\,g/mol} \times 1 \quad \therefore x = 12$

$C_xH_yO_z : H_2O = 1 : \frac{y}{2}$ より
(288)

$\frac{114\,g}{228\,g/mol} \times \frac{y}{2} = \frac{90\,g}{18\,g/mol} \times 1 \quad \therefore y = 20$

$C_{12}H_{20}O_z = 164 + 16z = 228 \quad \therefore z = 4$　　$C_{12}H_{20}O_4$

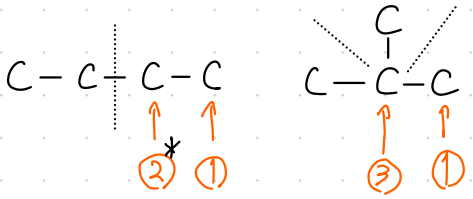
問2. A：$C_{12}H_{20}O_4$　$U=\frac{2\times12+2-20}{2}=3$　〜 C=C ×1 + −C(=O)−O− ×2

ジエステル A $C_{12}H_{20}O_4$ —けん化→ ㋐ アルコール B, C ($C_4H_{10}O$)

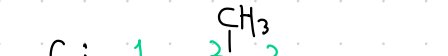
→ ㋑ D の Na 塩 —H^+→ D

* アルコール B —Ox→ アルデヒド　（第1級）
* アルコール C —Ox→ ✗　（第3級）
* B, C —$-H_2O$→ 同一アルケン　（同一C骨格）

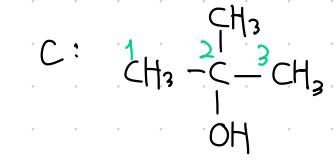
$C_4H_{10}O$ (U=0)



B：3-メチル-1-プロパノール



C：2-メチル-2-プロパノール



* D —$-H_2O$→ ✗
* シス・トランス
* E —$-H_2O$→ ○

D：トランス型
E：シス型

（C=C あり, −C(=O)−OH ×2）

U=3

ジエステル

A + $2H_2O$ ⟶ B + C + D

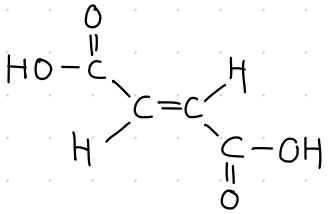
∴ D = A + $2H_2O$ − (B+C)

= $C_{12}H_{20}O_4$ + $2H_2O$ − 2× $C_4H_{10}O$

= $C_4H_4O_4$ (116)

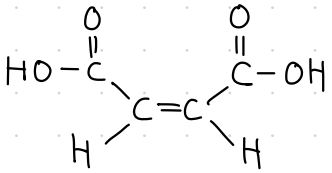
(U=3)

D：



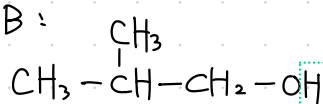
フマル酸

E：

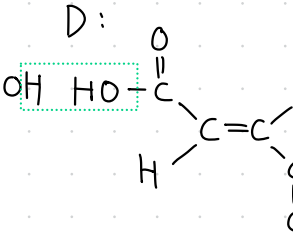


マレイン酸

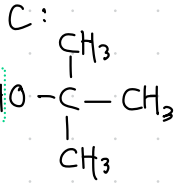
B：



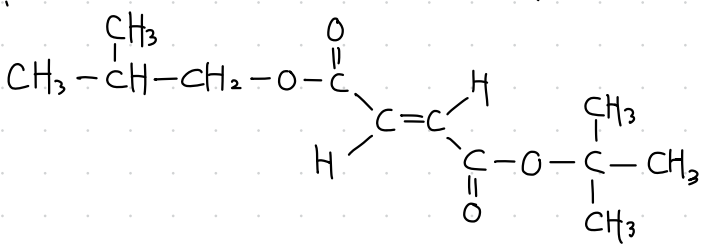
D：



C：



A：



3 - 7

問1　$C_7H_8O_4$

$$C:H:O = \frac{53.8}{12} : \frac{5.1}{1.0} : \frac{41.1}{16} = 4.48 : 5.1 : \underset{①}{2.56}$$
$$\fallingdotseq 1.75 : 1.99 : 1 = 7 : 8 : 4$$

$130 \leqq (C_7H_8O_4)_n \leqq 170 \quad \therefore n = 1.$　　$C_7H_8O_4$ (U=4)

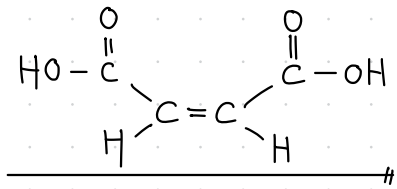
問2

酸性エステル A → けん化 → B の Na 塩 → H^+ → B（ジカルボン酸）

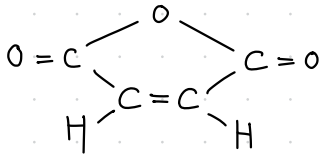
B：C=C あり／分子量 116 → $-H_2O$ → D

中性C：$-OH$ ✗、$-\overset{O}{\overset{\|}{C}}-H$ ✗、$I_2 + NaOH \rightarrow CHI_3\downarrow$

B：



D：



$A + H_2O \longrightarrow B + C$　　$\therefore C = A + H_2O - B$

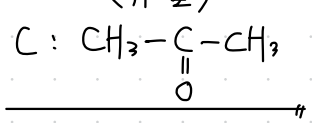
$= C_7H_8O_4 + H_2O - C_4H_4O_4$

$= C_3H_6O$　(U=1)

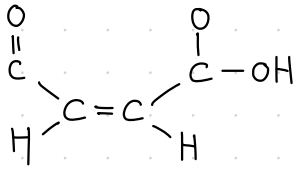
エステルの加水分解生成物として −OH 基を持つものがない
⟶ エノール型が ケト型に変わった.

(ケト型)

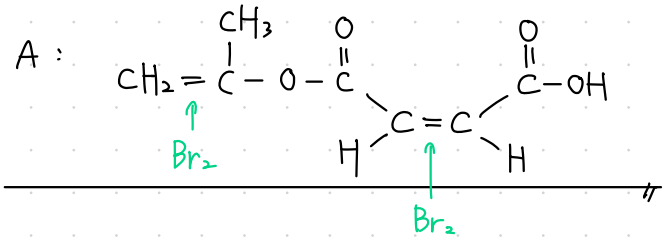
$\therefore C:$



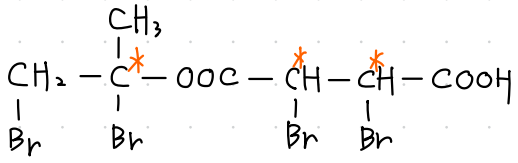
$CH_2=C(OH)-CH_3$



A：

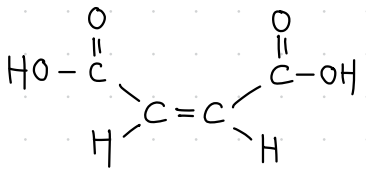


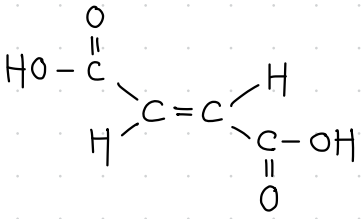
問3

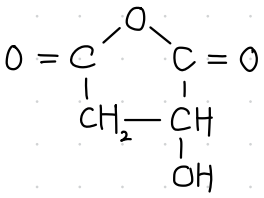


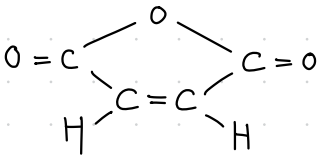
$2^3 = 8$

3－8

A 

B 

C 

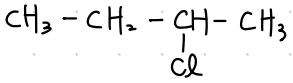
D 

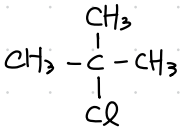
E $HOOC-CH_2-CH_2-COOH$

3－9

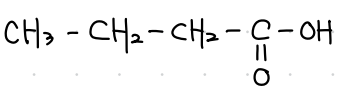
問1　4

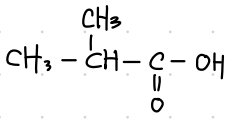
問2.



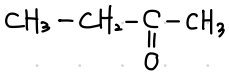


問3





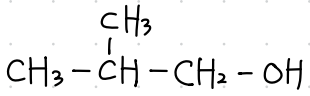
問4

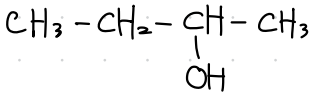


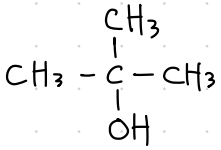
3.6 g

3－10

問1. A $CH_3-CH_2-CH_2-CH_2-OH$

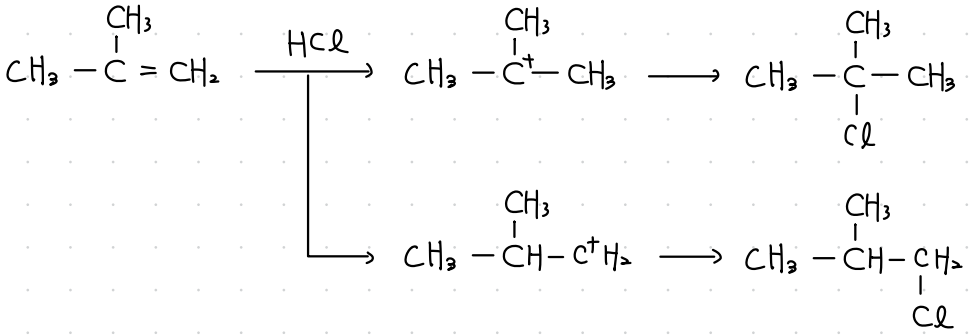
B 

C 

D 

E $CH_3-CH_2-O-CH_2-CH_3$

問2



問3 $CH_3-CH(OH)-CH_2-CH_3 + 4I_2 + 6NaOH \rightarrow CH_3-CH_2-C(=O)-ONa + CHI_3 + 5H_2O + 5NaI$

問4 C

問5 エーテル結合

3－11

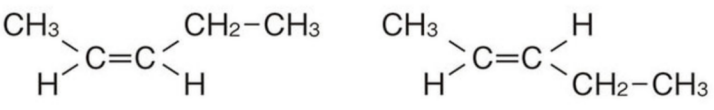
問1 A $CH_3-CH_2-CH(OH)-CH_2-CH_3$

B $CH_3-CH_2-C(CH_3)(OH)-CH_3$　C $CH_3-CH(CH_3)-CH_2-CH_2-OH$

D $CH_3-C(CH_3)_2-CH_2-OH$

F $CH_3-CH_2-CH_2-CH_2-CH_2-OH$

問2 K

問3 

3－12

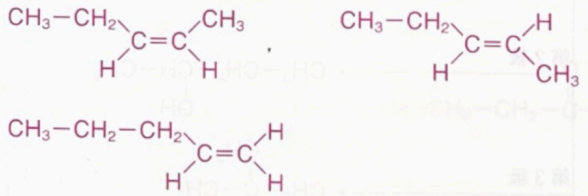
問1 A $CH_3-CH_2-CH_2-CH_2-CH_2-OH$

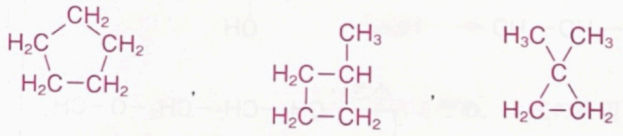
B $CH_3-CH_2-CH_2-\overset{*}{C}H(OH)-CH_3$

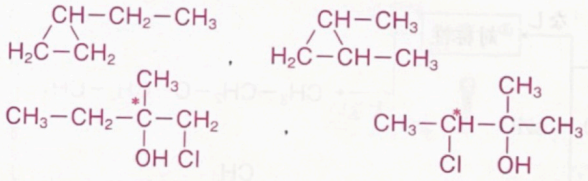
C $CH_3-CH_2-CH(OH)-CH_2-CH_3$　D $CH_3-CH_2-\overset{*}{C}H(CH_3)-CH_2OH$

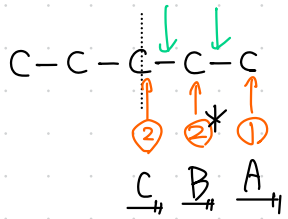
E $CH_3-CH_2-C(CH_3)(OH)-CH_3$　F $CH_3-\overset{*}{C}H(OH)-CH(CH_3)-CH_3$

問2 0.15 L　問3 銀鏡反応，~~アルデヒド基~~ ホルミル

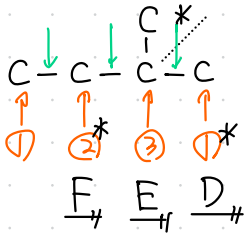
問4 

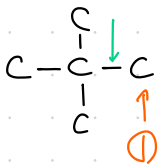
問5 

問6 



アルコール	① A	②* B	② C	①* D	③ E	②* F
		C直鎖		C分枝		
I_2+NaOH		+				+
$K_2Cr_2O_7$	了	ケ	ケ	了	×	ケ





問2 $2C_5H_{11}OH + 2Na \longrightarrow 2C_5H_{11}ONa + H_2$

(88)　(23)

アルコール：$\frac{10\,g}{88\,g/mol} = \frac{5}{44}\,mol$

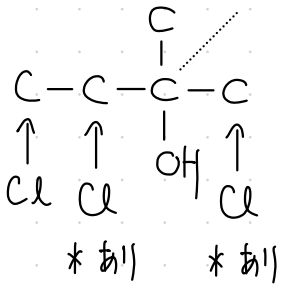
Na：$\frac{0.3\,g}{23\,g/mol} = \frac{3}{230}\,mol$

} 係数を割り算すると，アルコール過剰より Naに合わせて計算

Na : H_2 = 2 : 1 より

$\frac{0.3}{230}$ mol × $\frac{1}{2}$ × 22.4 L/mol = 0.15 L

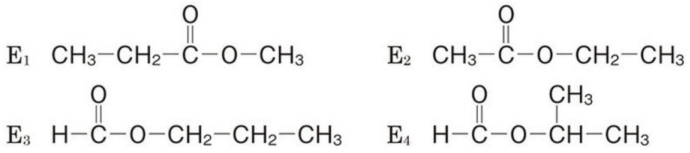
問6



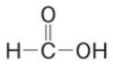
＊あり　＊あり

3－13

問1



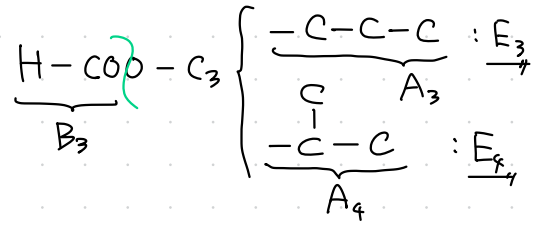
問2 ギ酸は分子内にホルミル基を持つため。(19字)

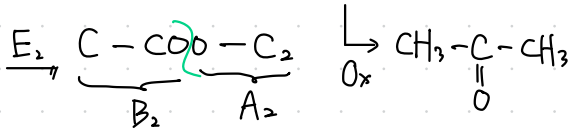


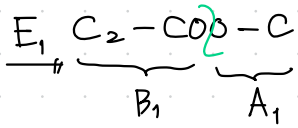
問3 酢酸カルシウム

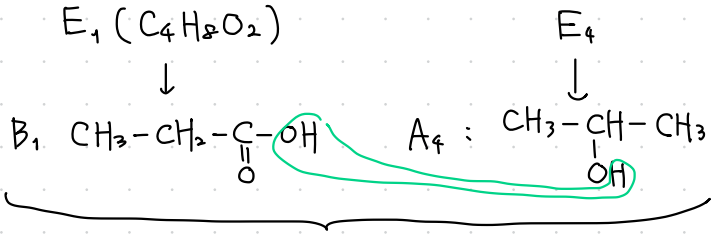
$(CH_3COO)_2Ca \rightarrow CH_3COCH_3 + CaCO_3$

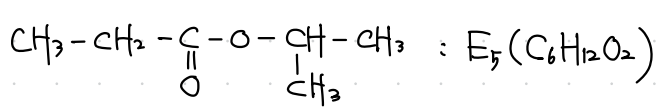
問4 4.4 g

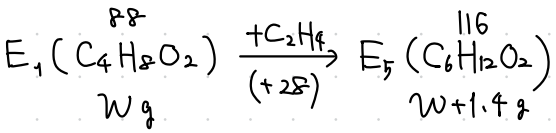








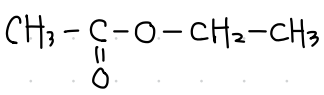


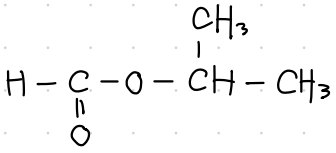


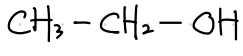
$$\frac{W\ g}{88\ g/mol} = \frac{W+1.4\ g}{116\ g/mol}$$

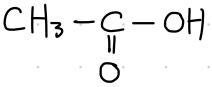
$$\therefore W = 4.4\ g$$

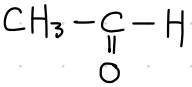
3－14

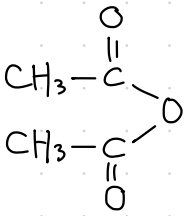
問1　A 

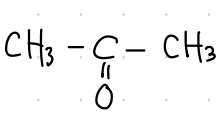
B 

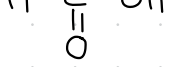
C 

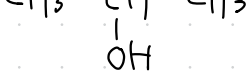
D 

E 

F 

G 

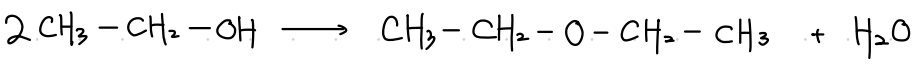
H 

I 

問2,3　略

問4.　アセトアルデヒド

問5　ジエチルエーテル



3－15

問1 $CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-OH + CH_3-OH \longrightarrow CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_3 + H_2O$

エステル化の触媒

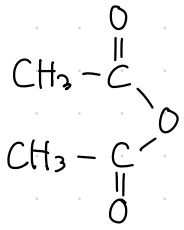
問2 $2C_5H_{10}O_2 + 13O_2 \rightarrow 10CO_2 + 10H_2O$ 1.46×10^{-1} L

問3 $CH_3-\underset{\underset{O}{\|}}{C}-O-CH_2-CH_2-CH_3$

B：酢酸

C：1－プロパノール

問4 $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$ $CH_3-CH_2-O-CH_2$

問5 

問6 ホルミル基

3－16

問1 10

問2 $H-\underset{\underset{O}{\|}}{C}-OH$, $CH_3-\underset{\underset{O}{\|}}{C}-OH$, $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-OH$,

$CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-OH$, $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{O}{\|}}{C}-OH$,

問3 ギ酸

問4 A $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$ B $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$

D $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH$

問5 $CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_3$, $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{O}{\|}}{C}-O-CH_3$

問6 $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_2-CH_3$

$C_5H_{10}O_2$　$U=1=-\overset{\overset{O}{\|}}{C}-O-\times 1$

$H-COO-C_4$ (C：ギ酸)
- $-C-C-C-C$ ①
- $-C-\overset{\overset{C}{|}}{C}-C$ ①
- $-\overset{\overset{C}{|}}{C^*}-C-C$ ②目 — B
- $-C-\overset{\overset{C}{|}}{C}-C$ ③

$C-COO-C_3$ (酢酸)
- $-C-C-C$ ① — A
- $-C-\overset{\overset{C}{|}}{C}-C$ ②目

$C_2-COO-C_2$ ①目

$C_3-COO-C$ ①
- $C-C-C-$
- $C-\overset{\overset{C}{|}}{C}-C-$

問5　$R-COO-R' \xrightarrow{+H_2O} RCOOH + R'OH$

$\underset{\overset{\|}{}}{RCOOH} + NaOH \longrightarrow \cdots$

$C_nH_{2n+1}COOH$ $(14n+46)$

$RCOOH : NaOH = 1:1$ より

$$\frac{528\times10^{-3}\,g}{14n+46\ g/mol} = 0.10\ mol/L \times \frac{60}{1000}\,L \qquad \therefore n=3$$

⟶ C_4 のカルボン酸のエステル　$CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-O-CH_3$　$CH_3-\underset{\underset{CH_3}{|}}{CH}-\overset{\overset{O}{\|}}{C}-O-CH_3$

問6　$R-COO-R' \xrightarrow{+H_2O} RCOOH + R'OH$

$\underset{\overset{\|}{}}{2R'-OH} + 2Na \longrightarrow 2R-ONa + H_2$

$C_nH_{2n+1}OH$ $(14n+18)$

$R'-OH : H_2 = 2:1$ より

$$\frac{230\times10^{-3}\,g}{14n+18\ g/mol}\times 1 = \frac{56\times10^{-3}\,L}{22.4\ L/mol}\times 2$$

$\therefore n=2$ ⟶ C_2 のアルコールのエステル　$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-O-CH_2-CH_3$